Chapter 19 worked solutions • Windows 7

OS19-E01 • Granted rights and SAME_ACCESS

1. Neither hasW in its granted mask. Writable parameter storage does not grant authority. Kernel operations check the required handle rights.
2. B's 0x40 can be invalid, refer to another object, or be an existing handle to F. Its number alone does not determine identity. The duplicated handle is B's 0x88; after A closes, that handle still references F, so F cannot yet be freed.
3. No. SAME_ACCESS preserves the mask. Different access depends on object/API/caller authority, including PROCESS_DUP_HANDLE for process access. Neither a raw number nor parent-child relation grants more rights. Prove the specified flag, not a universal duplication theorem.

OS19-E02 • Session scope

1. Per-process mappings, desktops/window stations, and selected scoped objects separate;kernel, drivers, physical resources, and possibly files/network remain shared. Profile location is not hardware isolation.
2. Local resolves in session namespace and can name different objects. Global may share identity but retains create/open/DACL checks. Handle numbers remain process-local.
3. No. Sessions share this kernel;VMs normally introduce guest-kernel/VMM boundaries. Driver faults/RAM pressure can affect multiple sessions. Identify scope and enforcement rather than merely different users.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

OS19-E03 • Actual ReadFile outcomes

1. Pending submission is not ordinary failure. Collect through GetOverlappedResult or the chosen path. Windows 7 requires a valid non-NULL DWORD count pointer;use final completion bytes for pending operations.
2. With the file unchanged and both reads successful, offset 0 returns 6 bytes (ABCDEF), and offset 6 returns 4 bytes (GHIJ). A positive-length overlapped read starting at offset 10 reports ERROR_HANDLE_EOF. ReadFile may report it immediately, or return ERROR_IO_PENDING first and let GetOverlappedResult report it later. Immediate EOF is not pending. Keep each OVERLAPPED, buffer, and event valid until its terminal result, then reuse it.
3. No. Cancellation is a request. Normal completion can win, abort can yieldERROR_OPERATION_ABORTED, andNOT_FOUND still requires knowing original result. Release only after terminal;shared handles/events survive all related operations. Submission, completion, cancel statuses differ.

OS19-E04 • IRQL and thread priority

1. Neither nonzero blocking wait nor a pageable access fault is legal here. Retain context and queue waiting/pageable work to an appropriate PASSIVE worker;keep DPC short/permitted.
2. No. IRQL is processor-local. Cross-CPU access needs suitable spinlock/other synchronization with correct IRQL/order rules.
3. ISR/DPC, nonpreemptible sections, drivers, and resource waits still affect response. Without bounds/worst-case analysis, priority 31 supplies no hard deadline. IRQL 2 is not thread priority 2.

OS19-E05 • Fixed priority queues

1. P1 runs 0–1;P3 runs 1–3/completes;P1 runs 3–4 with residual 1, leaving burst 2 at tail;P2 runs 4–6, leaves1;P1 runs 6–8/completes;P2 runs 8–9/completes. P1/P2/P3 finish 8/9/3.
2. It violates immediate higher-priority preemption. A nonpreemptive/delayed-preemption model needs changed premises and recomputation.
3. No. Real quantum, boost, decay depend on environment/workload. This fixed queue model is not a measurement of every heuristic. Track residual slice and queue order.

OS19-E06 • Reserve, commit, resident

1. Reserve 64 KiB, commit 8 KiB, resident 0. The specified successful first touch demand-zeroes the first page, giving 4 KiB resident while other quantities remain. No other page loads or evictions occur.
2. Decommit removes first-page commitment/residency, leaving 4 KiB committed, 0 resident, and 64 KiB reserved. Release removes the reservation, all0. Actual release base/size rules still apply.
3. No. Commit promises backing resources;physical allocation can await touch. It does not immediately write pagefile pages. Equal process VAs also do not compete globally. These are different sets/statistics.

OS19-E07 • Ordered DACL

1. Remaining 3→allow Alice removes 1→remaining 2. Deny Staff intersects 2 and stops with denial; the final allow is unreachable.
2. After first allow leavesW, earlier allowStaff fills it and succeeds before deny. Order matters;canonical ordering is preferred rather than a universal deny-overrides rule.
3. NULL imposes no DACL restriction;empty exists without a granting ACE and denies this ordinary request. Other token/policy layers remain outside the simplified model. Use remaining rather than arbitrary entry union.

OS19-E08 • NTFS extents

1. VCN 5, offset 123 selects LCN 201. Volume 823419, disk 1871995. LCN is volume-relative, so include partition base.
2. Hole returns zeros, not LCN 0. Eleven logical clusters=45056 B; nine data clusters=36864 B without compression/metadata.
3. Resident means attribute content in MFT record, not RAM residency.48-bit record/16-bit sequence rejects mismatched reused references;finite sequence does not prove no wrap. Preserve layout/units.

OS19-E09 • Single-transaction WAL

1. Undo old values: bit=0, map[100], size 4096. Do not retain bit=1 while declaring transaction absent. The complete old/new log and marker are durable before any home change.
2. Redo all new values: bit=1, map[100,200], size 8192. Required new log precedes durable commit, so home lag is recoverable.
3. No. Retain active undo/dirty redo until safe truncation;elapsed seconds alone do not authorize deletion. Metadata consistency is not all user-data or cross-file business consistency. Provide application persistence/publication. This proves the single specified transaction model, not complete NTFS concurrent recovery.

OS19-E10 • Snapshots and remote uncertainty

1. Preserve old 41 before first change;S stays 41 while live 42 then 43. Unchanged Y shares 80. Reclaim backing only after all snapshots/readers release it.
2. No. Writer coordination can be required for application-consistent multi-field state. Shared storage failure can destroy both, so it is not independent backup.
3. No. Server may append before losing reply. Request identity, deduplication, idempotence, or protocol retry rules are required. TCP bytes alone do not make append exactly-once. Separate shared versions and fault domains.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 19. Microsoft: IRQL, VirtualAlloc, AccessCheck. ReadFile, EOF, CancelIoEx, named pipes.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.

https://github.com/xinhi6716/computer-science-self-study